Signal and System Design (IGS2137)

Week 11 Lab

Jump Together, Fly Farther!

School of Global Convergence Studies

인하대학교 국제학부

ISE Department

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Midterm Review Exercise:

Goals :

1. Repeat the same operations (shifting, scaling, reversing)
2. Write a report explaining each step of the implementation
3. Understand and implement algorithms with MATLAB

Midterm Review Exercise:

Task:

- Make a program to perform x3 operations as shown below:

- The user should input the value as:
  a) Type of operation(shifting, scaling, reversing)
  b) Type of Signal (Discrete or Continuous)
  c) Amount of shift or value based on the type of operation user selected

Midterm Review Exercise:

Assumptions:

- Amount of shift/scaling/reversing can be + or -
- Exception should be thrown in case the user input an unrealistic value

Challenge:

- You need to select 4 different signals instead of the 3 used in lecture material
- One must of sinusoidal wave (sine or cosine)
- One must be Saw tooth or Square signal
- Other two can be any of your choices
- Implementing using App Designer

Exercise 1:

4.56 The rectangular window is defined as

$$w_r[n] = \begin{cases} 1, & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}.$$

We may truncate the duration of a signal to the interval $0 \le n \le M$ by multiplying the signal with $w[n]$. In the frequency domain, we convolve the DTFT of the signal with

$$W_r(e^{j\Omega}) = e^{-j\frac{M}{2}\Omega} \frac{\sin\left(\frac{\Omega(M+1)}{2}\right)}{\sin\left(\frac{\Omega}{2}\right)}.$$

The effect of this convolution is to smear detail and introduce ripples in the vicinity of discontinuities. The smearing is proportional to the mainlobe width, while the ripple is proportional to the size of the sidelobes. A variety of alternative windows are used in practice to reduce sidelobe height in return for increased mainlobe width. In this problem, we evaluate the effect of windowing time-domain signals on their DTFT. The role of windowing in filter design is explored in Chapter 8.

The Hanning window is defined as

$$w_h[n] = \begin{cases} 0.5 - 0.5\cos\left(\frac{2\pi n}{M}\right), & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$$

(a) Assume that $M = 50$ and use the MATLAl command `fft` to evaluate the magnitude spec trum of the rectangular window in dB at inter vals of $\frac{\pi}{50}$, $\frac{\pi}{100}$, and $\frac{\pi}{200}$.

(b) Assume that $M = 50$ and use the MATLAl command `fft` to evaluate the magnitude spec trum of the Hanning window in dB at interval of $\frac{\pi}{50}$, $\frac{\pi}{100}$, and $\frac{\pi}{200}$.

(c) Use the results from (a) and (b) to evaluate th mainlobe width and peak sidelobe height in dl for each window.

(d) Let $y_r[n] = x[n]w_r[n]$ and $y_h[n] = x[n]w_h[n]$ where $x[n] = \cos\left(\frac{26\pi}{100}n\right) + \cos\left(\frac{29\pi}{100}n\right)$ an $M = 50$. Use the the MATLAB command `ff` to evaluate $|Y_r(e^{j\Omega})|$ in dB and $|Y_h(e^{j\Omega})|$ in dB a intervals of $\frac{\pi}{200}$. Does the choice of window af fect whether you can identify the presence o two sinusoids? Why?

(e) Let $y_r[n] = x[n]w_r[n]$ and $y_h[n] = x[n]w_h[n]$ where $x[n] = \cos\left(\frac{26\pi}{100}n\right) + 0.02\cos\left(\frac{51\pi}{100}n\right)$ an $M = 50$. Use the the MATLAB command `ff` to evaluate $|Y_r(e^{j\Omega})|$ in dB and $|Y_h(e^{j\Omega})|$ in dB at intervals of $\frac{\pi}{200}$. Does the choice of window affect whether you can identify the presence of two sinusoids? Why?

Exercise 1:

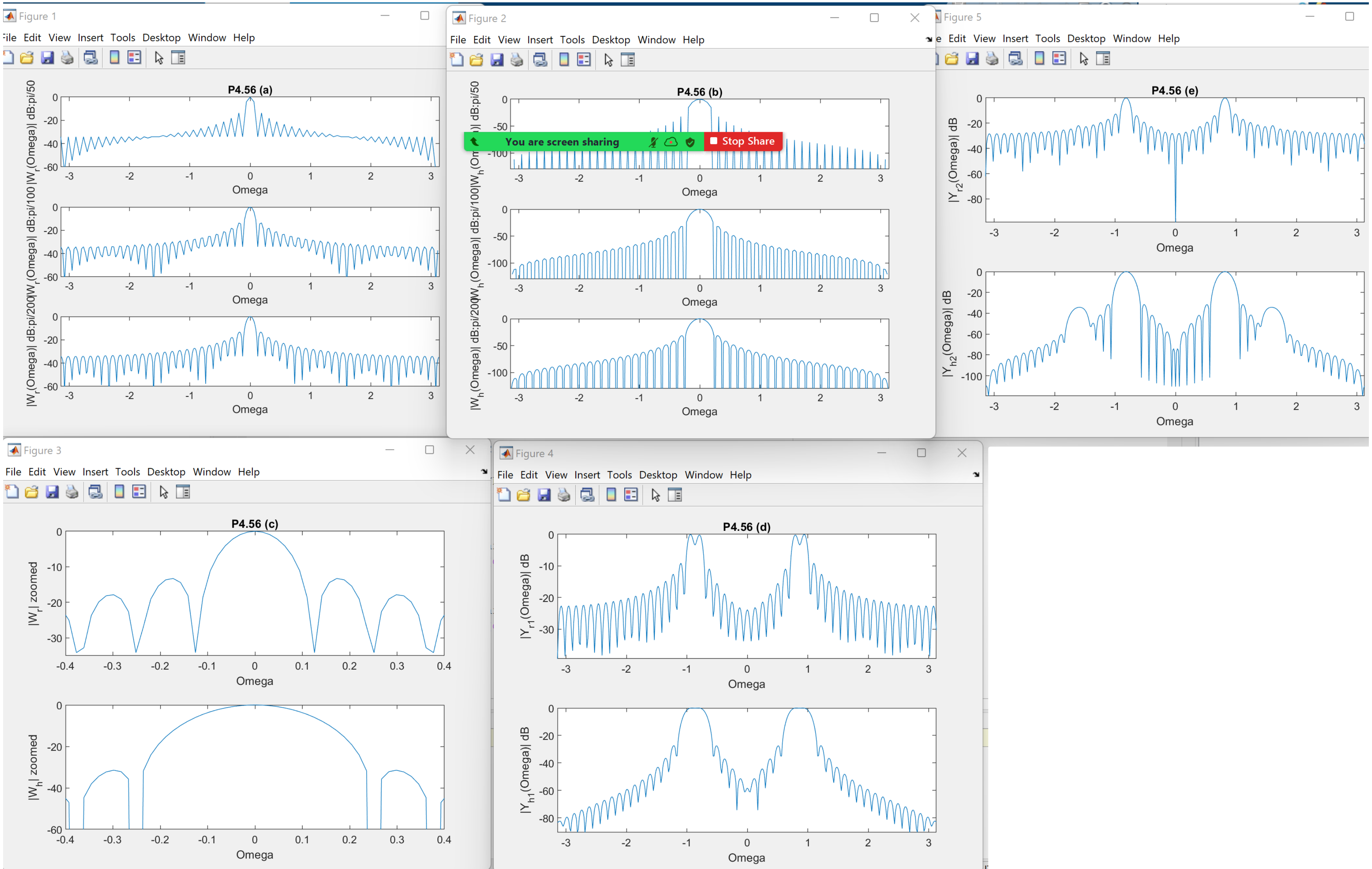


Figure 1
P4.56 (a)
Omega
Figure 2
P4.56 (b)
You are screen sharing
Stop Share
Omega
Figure 5
P4.56 (e)
Omega
Figure 3
P4.56 (c)
Omega
Figure 4
P4.56 (d)
Omega

Exercise 2:

4.57 Let the discrete-time signal

$$x[n] = \begin{cases} e^{-\frac{(0.1n)^2}{2}}, & |n| \le 50 \\ 0, & \text{otherwise} \end{cases}.$$

Use the MATLAB commands `fft` and `fftshift` to numerically evaluate and plot the DTFT of $x[n]$ and the following subsampled signals at 500 values of Ω on the interval $-\pi < \Omega \le \pi$:

(a) $y[n] = x[2n]$

(b) $z[n] = x[4n]$

Exercise 3:

4.57 Let the discrete-time signal

$$x[n] = \begin{cases} e^{-\frac{(0.1n)^2}{2}}, & |n| \le 50 \\ 0, & \text{otherwise} \end{cases}.$$

Use the MATLAB commands `fft` and `fftshift` to numerically evaluate and plot the DTFT of $x[n]$ and the following subsampled signals at 500 values of Ω on the interval $-\pi < \Omega \le \pi$:

(a) $y[n] = x[2n]$

(b) $z[n] = x[4n]$

4.58 Repeat Problem 4.57, assuming that

$$x[n] = \begin{cases} \cos\left(\frac{\pi}{2}n\right)e^{-\frac{(0.1n)^2}{2}}, & |n| \le 50 \\ 0, & \text{otherwise} \end{cases}.$$

Exercise 4:

4.59 A signal is defined as

$$x(t) = \cos\left(\frac{3\pi}{2}t\right)e^{-\frac{t^2}{2}}.$$

(a) Evaluate the FT $X(j\omega)$, and show that $|X(j\omega)| \approx 0$ for $|\omega| > 3\pi$.

In parts (b)–(d), we compare $X(j\omega)$ with the FT of the sampled signal, $x[n] = x(nT_s)$, for several sampling intervals. Let $x[n] \xleftrightarrow{FT} X_\delta(j\omega)$ be the FT of the sampled version of $x(t)$. Use MATLAB to numerically determine $X_\delta(j\omega)$ by evaluating

$$X_\delta(j\omega) = \sum_{n=-25}^{25} x[n]e^{-j\omega T_s}$$

at 500 values of ω on the interval $-3\pi < \omega < 3\pi$. In each case, compare $X(j\omega)$ and $X_\delta(j\omega)$ and explain any differences.

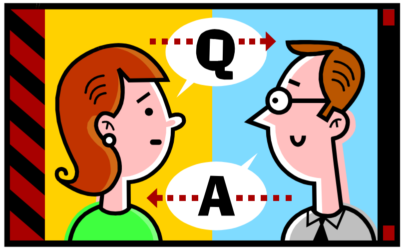
(b) $T_s = \frac{1}{3}$

(c) $T_s = \frac{2}{5}$

(d) $T_s = \frac{1}{2}$

Brief break (if on schedule)

Q&A?



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